

Synthesis and stability study of a new major metabolite of γ -hydroxybutyric acid

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Summary γ -Hydroxybutanoic acid (GHB) is used as a date-rape drug, which renders the victims unconscious and defenceless. Intoxications are very difficult to detect for forensic scientists due to rapid metabolism to endogenous levels of GHB. We recently discovered a new major metabolite, 2, of GHB (1) that could potentially extend the analytical detection window for GHB intoxications. Herein we disclose synthetic procedures based on a Koenigs–Knorr glucuronidation approach that provides GHB glucuronide 2 and a deuterium-labelled analogue d 4-2 of high purity suitable for analytical chemistry. In addition, we have assessed the stability of GHB glucuronide 2 by mimicking the natural pH range for urine, which is of importance in the development of new analytical methods. Using NMR we show that GHB glucuronide 2 is highly stable towards aqueous hydrolysis within the pH range normally observed for urine even at elevated temperature.